Carbon and fossil fuels

CARBON AND ITS COMPOUNDS FORM THE BASIS FOR ALL FOSSIL FUELS.

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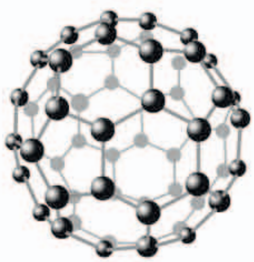
After hydrogen, carbon is the most common element in living things. When organisms die, their remains are preserved underground. Over millions of years are transformed into useful, carbon-rich compounds called fossil fuels.

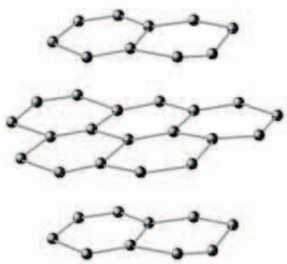
Forms of carbon

Pure carbon exists in different forms, or allotropes (see page 111). The carbon atoms in each allotrope link up in different ways, which gives the allotropes very different properties. Diamond is an extremely hard and sparkling gem. The arrangement of the atoms in graphite, however, make it a slippery gray solid, often used as pencil lead.

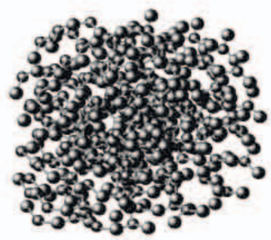
△ **Diamond**
The carbon atoms are arranged in a very rigid crystal network based on repeating tetrahedra of four atoms.



△ **Fullerene**
The atoms link together to form a ball-shaped cage. Fullerenes may contain 100, 80, or 60 carbon atoms.



△ **Graphite**
The atoms are arranged in sheets of hexagons. The sheets are only loosely bonded, so they slip over each other.



△ **Soot**
The atoms in this allotrope are arranged randomly. Soot forms from the uncombusted carbon released when fossil fuels burn.

Coal

Coal is a carbon-rich sedimentary rock made from the remains of trees. Most of the coal mined today formed from forests that grew around 300 million years ago. The plant material was buried in the absence of oxygen, so huge amounts were preserved as sediments, gradually forming coal.

▷ **Coal formation**
The process begins when plant remains sink in waterlogged, boggy soil. The lack of oxygen prevents the wood from decaying. These remains form a dense soil called peat, which can itself be used as a fuel when dried. Over time the peat is buried, and the increased pressure drives out the water to form lignite (soft, brown rock). Deeper down, heat hardens the lignite into coal.